

- 1) Bob was doing some hiking in order to clear his mind after last week's AERSP 309 homework when he looked up and noticed a UFO hovering 80 km directly above him, holding its position steady with respect to the ground (i.e., co-rotating with the Earth). Bob quickly pulled out his notebook and his GPS device and recorded his coordinates as $(\phi, \lambda) = (40.8^\circ, 282.1^\circ)$ (a.k.a. 40.8° N, 77.9° W — basically State College). Assuming a spherical Earth with radius $R_\oplus = 6370$ km that rotates with angular velocity $\vec{\omega}_{\mathcal{E}/\mathcal{N}} \approx 7.29 \times 10^{-5}$ rad/s and $\theta = 35^\circ$ (see Fig. 1), please find the inertial position, velocity, and acceleration of the UFO with its coordinates expressed in the $\mathcal{N}$ -frame and $\mathcal{T}$ -frame, at the time of the sighting. [This extends the topocentric setup from HW1 P5 to velocity and acceleration via the transport theorem. Note that even though the UFO is stationary in your $\mathcal{T}$ -frame, its *inertial* velocity is nonzero because $\mathcal{T}$ itself rotates with the Earth.]

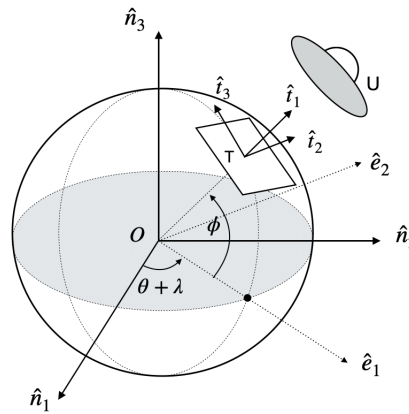


Fig. 1. Schematic of UFO sighting.

- 2) A constantly rotating disk is attached to a spacecraft. The spacecraft is moving with varying velocity $v(t)$ in a fixed direction. A particle, P , is attached to the disk at a distance of r from its center. Take the $\mathcal{S}$ -frame to be the spacecraft body frame with origin at the disk center, $\hat{s}_1$ along the direction of spacecraft motion, and $\hat{s}_3$ perpendicular to the disk (out of the page in Fig. 2); so the disk spins about $\hat{s}_3$ with $\vec{\omega} = \omega \hat{s}_3$. The particle sits at $\vec{r} = r \hat{d}_1$, where $\hat{d}_1$ is a disk-fixed unit vector that coincides with $\hat{s}_1$ at $t = 0$. For parts (a)–(c), you may treat the spacecraft's translational motion as decoupled from the rotational analysis (i.e., focus on the disk's contribution to the inertial velocity/acceleration in $\mathcal{S}$ -frame components).
- Assuming r and ω are constant, what is the inertial velocity and acceleration of the particle expressed in the $\mathcal{S}$ -frame?
 - Assuming ω is constant but r is now varying, what is the inertial velocity and acceleration of the particle expressed in the $\mathcal{S}$ -frame?
 - Assuming r is constant but ω is now varying, what is the inertial velocity and acceleration of the particle expressed in the $\mathcal{S}$ -frame?
 - Now assume both $r(t)$ and $\omega(t)$ vary. Write the full inertial acceleration of the particle in $\mathcal{S}$ -frame components and identify each term (centripetal, Coriolis, tangential/angular, radial/extension). Confirm that setting $\dot{r} = \ddot{r} = 0$ and $\dot{\omega} = 0$ recovers your answer to part (a).
- 3) Consider a simplified planetary system consisting of the Sun, Earth, and Moon as shown in Fig. 3. The Sun is fixed in the inertial $\mathcal{S}$ -frame ($\mathcal{S} : \{S, \hat{s}_1, \hat{s}_2, \hat{s}_3\}$), while the Earth and its $\mathcal{E}$ -frame ($\mathcal{E} : \{E, \hat{e}_r, \hat{e}_\alpha, \hat{e}_3\}$) orbit the Sun in a circular manner with constant rate $\dot{\alpha}$. The Moon and its $\mathcal{M}$ -frame ($\mathcal{M} : \{M, \hat{m}_r, \hat{m}_\beta, \hat{m}_3\}$) orbit the Earth in circular manner with constant rate $\dot{\beta}$. During a fairly routine mission to the Moon, an astronaut named Fred has been left behind in a circular orbit around the Sun with rate $\dot{\gamma}$, along with his $\mathcal{F}$ -frame (unlucky!). Assume all four orbits are coplanar (in the $\hat{s}_1$ – $\hat{s}_2$ plane), so $\hat{s}_3 = \hat{e}_3 = \hat{m}_3 = \hat{f}_3$ and all angular-velocity vectors point along $\hat{s}_3$. Let R_E be the Sun–Earth distance, r_M the Earth–Moon distance,

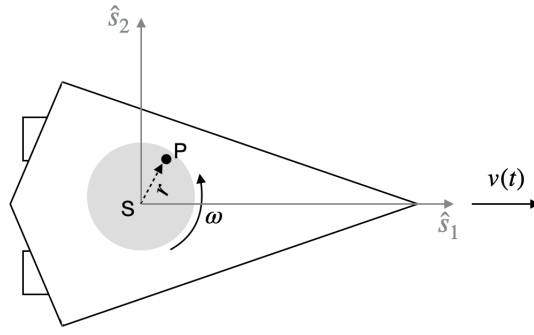


Fig. 2. Rotating disk on spacecraft.

and R_F Fred's orbital radius (all constant). You can help him sort out this suboptimal predicament by doing the following:

- Find the inertial velocity and acceleration of the Moon with respect to the Sun.
- Find the position vector of the Moon relative to the Earth.
- Find the angular velocity vectors $\vec{\omega}_{\mathcal{E}/\mathcal{F}}$ and $\vec{\omega}_{\mathcal{M}/\mathcal{F}}$.
- Find the velocity and acceleration vectors of the Moon as seen in Fred's frame.
- Fred is really hoping the Moon will appear to slow down and pause overhead so his rescue can catch up. Derive a general condition (in terms of $\dot{\alpha}, \dot{\beta}, \dot{\gamma}, R_E, r_M, R_F$, and the instantaneous geometry) under which the Moon's velocity in Fred's frame is exactly zero. Comment briefly on whether such a moment ever actually occurs for realistic values ($\dot{\gamma}$ close to $\dot{\alpha}$, $r_M \ll R_E \approx R_F$).

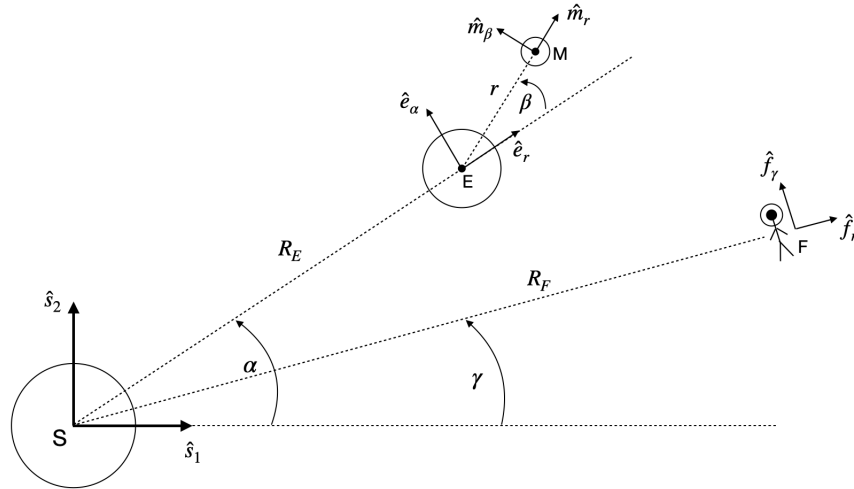
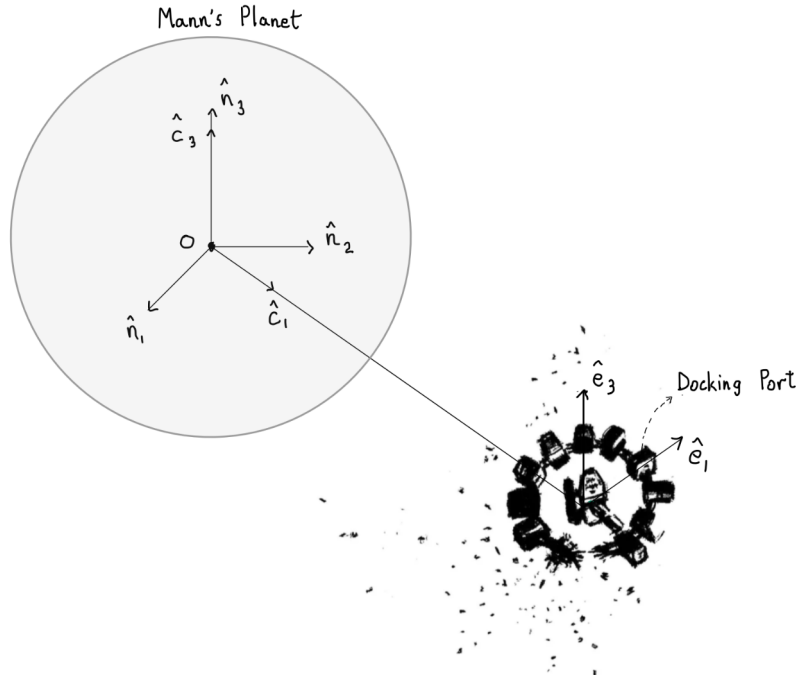


Fig. 3. Schematic of Sun-Earth-Moon-Fred motion.

- No time for caution! In the movie *Interstellar*, the Endurance space station is spinning rapidly after being damaged while in orbit around Mann's planet. Cooper and the crew attempt to dock their lander to the spinning station by matching its rotation. The Endurance is rotating about its central axis $\hat{e}_3$ (aligned with the inertial $\hat{n}_3$ vector) at a constant angular velocity of

$$\vec{\omega}^{\mathcal{E}/\mathcal{C}} = 0.65 \hat{n}_3 \text{ rad s}^{-1}$$

At initial time $t_0 = 0$, Cooper's lander is aligned with the inertial frame, and is preparing to align itself with the rotating $\mathcal{E}$ frame attached to the Endurance. At a certain time $t_1 = 6 \text{ s}$, the lander's onboard computer must compute the rotation required to align itself with the rotating frame $\mathcal{E}$ of the Endurance to initiate docking.



In addition to the rotation, the Endurance is also in a circular orbit in the equatorial plane of Mann's planet (i.e., orbit is in the plane defined by $\hat{n}_1$ and $\hat{n}_2$). Endurance's center of mass is at a radius of 5000 km from the center of Mann's planet along the $\hat{c}_1$ vector and has a constant orbital angular velocity given by

$$\vec{\omega}^{C/N} = 1.7857 \times 10^{-3} \hat{c}_3 \text{ rad s}^{-1}$$

- Compute the DCM that maps from the $\mathcal{N}$ -frame to the $\mathcal{E}$ frame at time t_1 .
 - A docking port is located at a fixed position 10 m away from the center of the Endurance station along the $\hat{e}_1$ direction. Express the position vector of the docking port at time t_1 in terms of components of the $\mathcal{C}$ -frame. Assume that $\hat{c}_1$, $\hat{e}_1$ and $\hat{n}_1$ are all aligned at the initial time t_0 .
 - Compute the inertial velocity vector of the docking port and express in terms of the $\hat{c}_i$ vectors.
- The angular velocity vectors of a spacecraft ($\mathcal{S} : \{\hat{S}, \hat{s}_1, \hat{s}_2, \hat{s}_3\}$) and a reference frame motion ($\mathcal{R} : \{\hat{R}, \hat{r}_1, \hat{r}_2, \hat{r}_3\}$) relative to the inertial frame are given by $\vec{\omega}_{S/N}$ and $\vec{\omega}_{R/N}$. The vector $\vec{\omega}_{S/N}$ is given in $\mathcal{S}$ -frame components and $\vec{\omega}_{R/N}$ is given in $\mathcal{R}$ -frame components. The error angular velocity vector of the spacecraft relative to the reference motion is denoted $\delta\vec{\omega} = \vec{\omega}_{S/N} - \vec{\omega}_{R/N}$. Find the relative error angular acceleration $\delta\dot{\vec{\omega}}$ with respect to an observer in the inertial frame, but its components expressed in the $\mathcal{S}$ -frame.
 - Find the angular velocity vector for the following three scenarios, and give two expressions for it: one in terms of the $\mathcal{B}$ frame components and the other in terms of the $\mathcal{N}$ frame components. In all cases, the Euler angle rates are $(\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3) = (1 \text{ deg/s}, 2 \text{ deg/s}, 3 \text{ deg/s})$
 - 3-2-3 Euler angles: $(\theta_1, \theta_2, \theta_3) = (10, -15, 30)$
 - 3-2-3 Euler angles: $(\theta_1, \theta_2, \theta_3) = (10, 0, 30)$
 - 1-2-3 Euler angles: $(\theta_1, \theta_2, \theta_3) = (10, 0, 30)$

Let's say there is a rotating frame, $\mathcal{B}$, undergoing small departure rotations relative to a reference frame, $\mathcal{R}$. Which set of Euler angles (3-2-3 or 1-2-3) is better suited for describing the motion of the $\mathcal{B}$ -frame?

- Fun Thought Problem:** Let's derive a part of the transport theorem. Let the $\mathcal{B}$ -frame ($\mathcal{B} : \{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$) rotate with an angular velocity of $\vec{\omega}_{B/N} = \omega \hat{b}_3$ with respect to the inertial $\mathcal{N}$ -frame ($\mathcal{N} : \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$).
 - The basis vectors of the $\mathcal{B}$ -frame at any instant in time, t , are related to the basis vectors at time $t = 0$

through a rotation

$$\hat{b}_i(t) = R\hat{b}_i(0) \quad \text{where} \quad R = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Differentiating with respect to time yields $\dot{\hat{b}}_i = \dot{R}\hat{b}_i(0)$. Find $\dot{R}$.

- b) Show that by plugging in the equality $R^T R = I_{3 \times 3}$, one can obtain the relation $\dot{\hat{b}}_i(t) = \dot{R}R^T \hat{b}_i(t)$.
- c) Show that $\dot{R}R^T$ is skew-symmetric (i.e., $\dot{R}R^T = -(\dot{R}R^T)^T$).
- d) Any skew-symmetric matrix can be replaced by the cross product of a vector

$$\text{i.e.,} \quad \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} = [\vec{a} \times].$$

What would the vector $\vec{a}$ have to be in the case of $\dot{R}R^T$? What then does the expression for $\dot{\hat{b}}(t)$ become?

- e) Recall that we have assumed a particular form of $\vec{\omega}_{B/N}$ at the beginning of this derivation. How could these steps be generalized to show that this result holds for any angular velocity vector?